

# TuniCars+

## Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

20260813\_134726\_6487d2

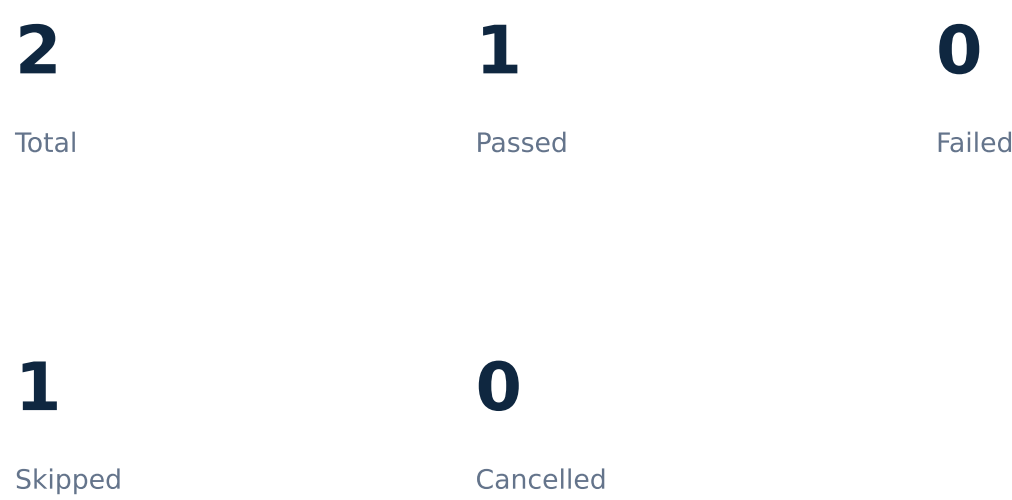
Date

2026-08-13T12:47:26.037980+00:00

# Executive Summary

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Overall measured availability: 16.67%

During this campaign, 1 of 2 scenarios passed, 0 failed and 1 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

# Test Environment

Architecture and technologies

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## Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

# Test Scenarios

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| Scenario       | swarm |
|----------------|-------|
| container-kill | 1     |
| node-failure   | 1     |

# Full Results Table

| Scenario       | Platform | Detection | Recovery | Required | Success | Avg ms       | P95 ms       | HTTP Fail | Avail. %     | Status  | Skip Reason                                                         | Error Message |
|----------------|----------|-----------|----------|----------|---------|--------------|--------------|-----------|--------------|---------|---------------------------------------------------------------------|---------------|
| container-kill | swarm    | 1.969     | 11.344   | True     | True    | 8.122        | 28.923       | 10        | 16.67        | PASS    | N/A                                                                 | N/A           |
| node-failure   | swarm    | N/A       | N/A      | N/A      | N/A     | Not Measured | Not Measured | N/A       | Not Measured | SKIPPED | Single-node Docker Swarm cannot demonstrate real node rescheduling. | N/A           |

# swarm — container-kill

State: **Pass**  
Detection Time (s): 1.969  
Recovery Time (s): 11.344

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HTTP Success: N/A

HTTP Failures: 10

Availability (%): 16.67

CPU (%): N/A

Memory (MB): N/A

## Recorded interpretation

No error or limitation recorded.

# swarm — node-failure

State: SKIPPED  
Detection Time (s): N/A

Recovery Time (s): N/A

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HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

## Recorded interpretation

Single-node Docker Swarm cannot demonstrate real node rescheduling.

# Availability

Current-session measured data

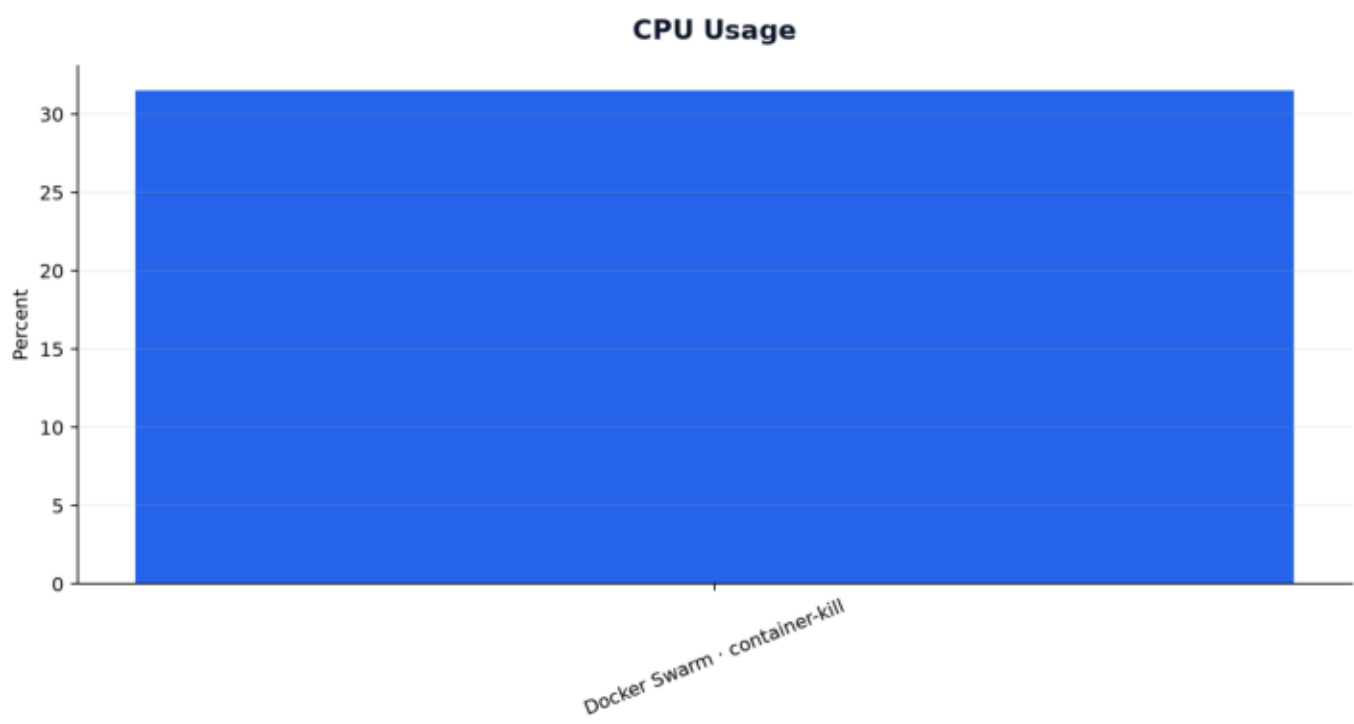
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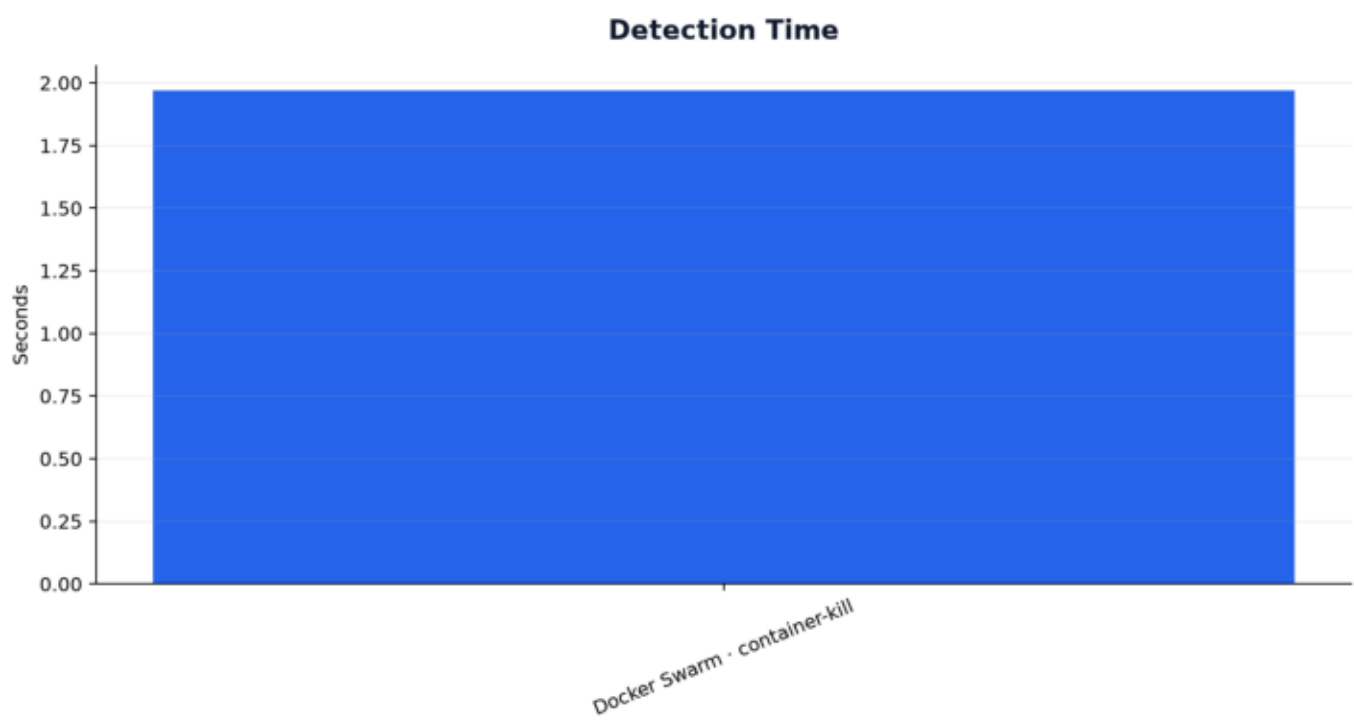
# Cpu Usage

Current-session measured data



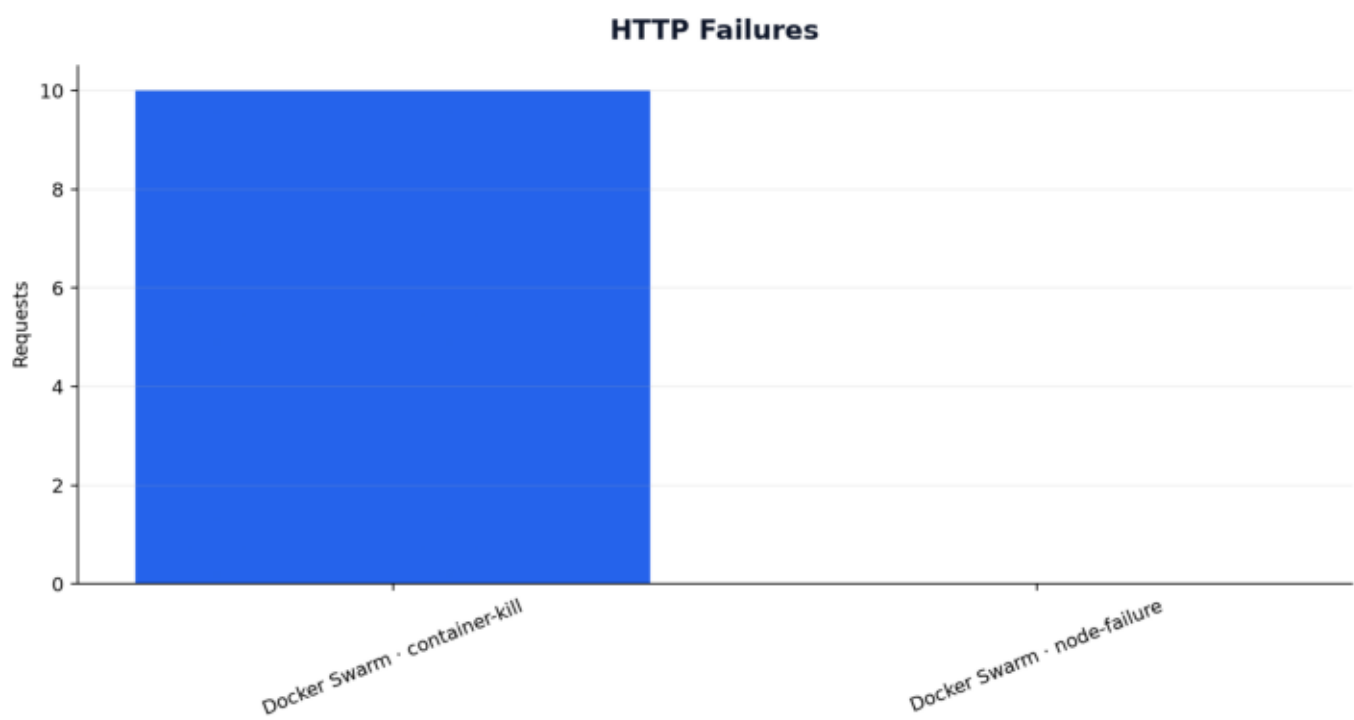
# Detection Time

Current-session measured data



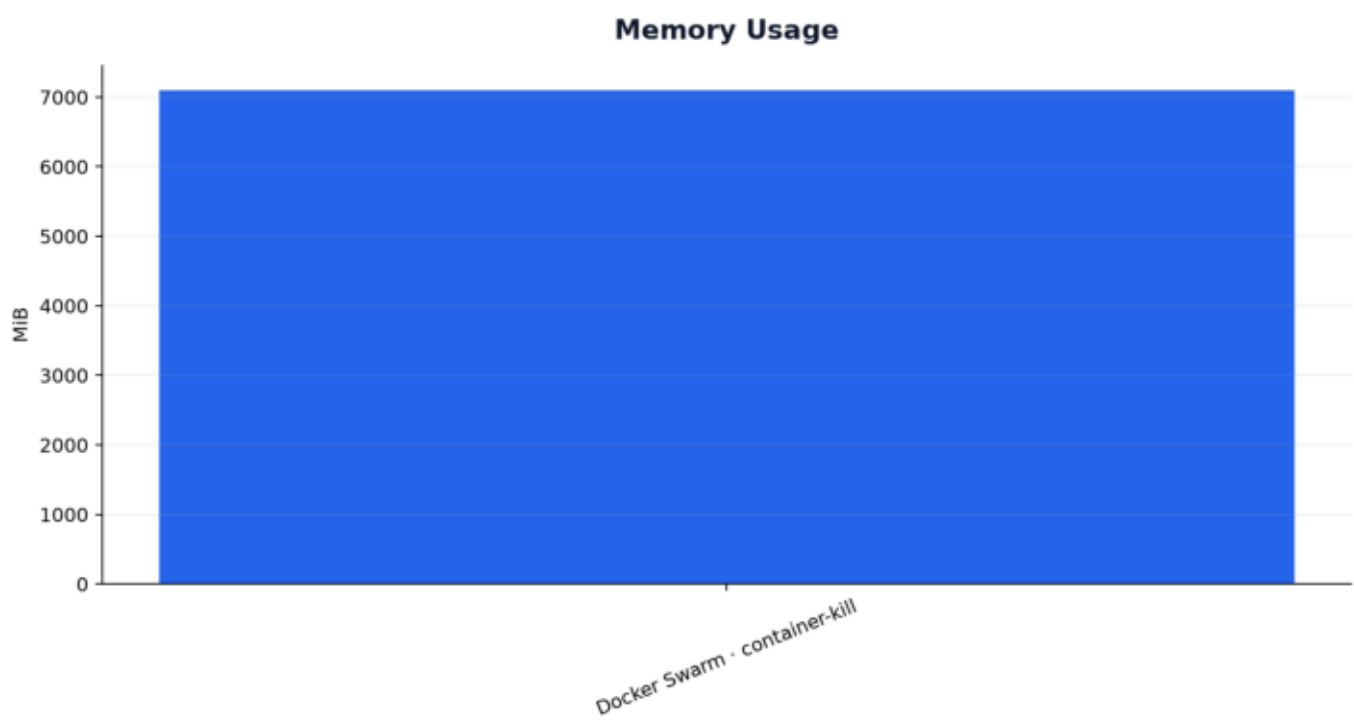
# Http Failures

Current-session measured data



# Memory Usage

Current-session measured data



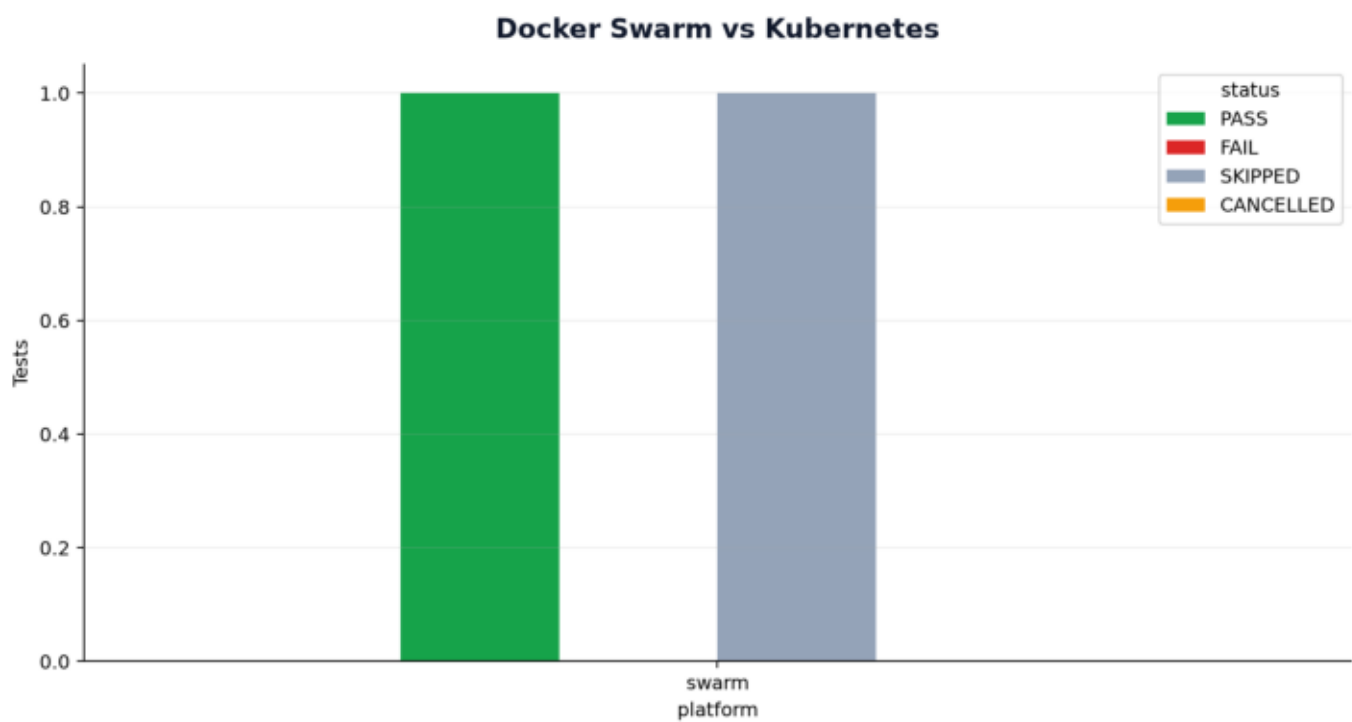
# P95 Response Time

Current-session measured data



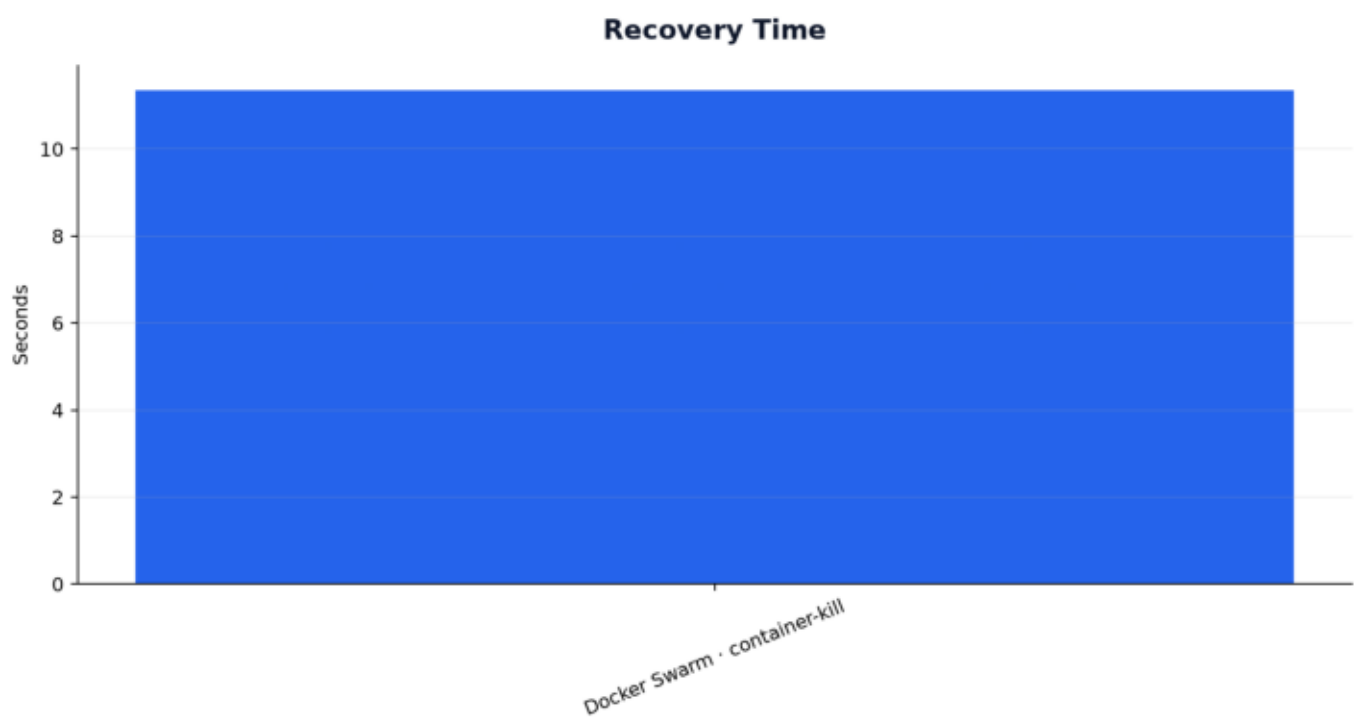
# Platform Comparison

Current-session measured data



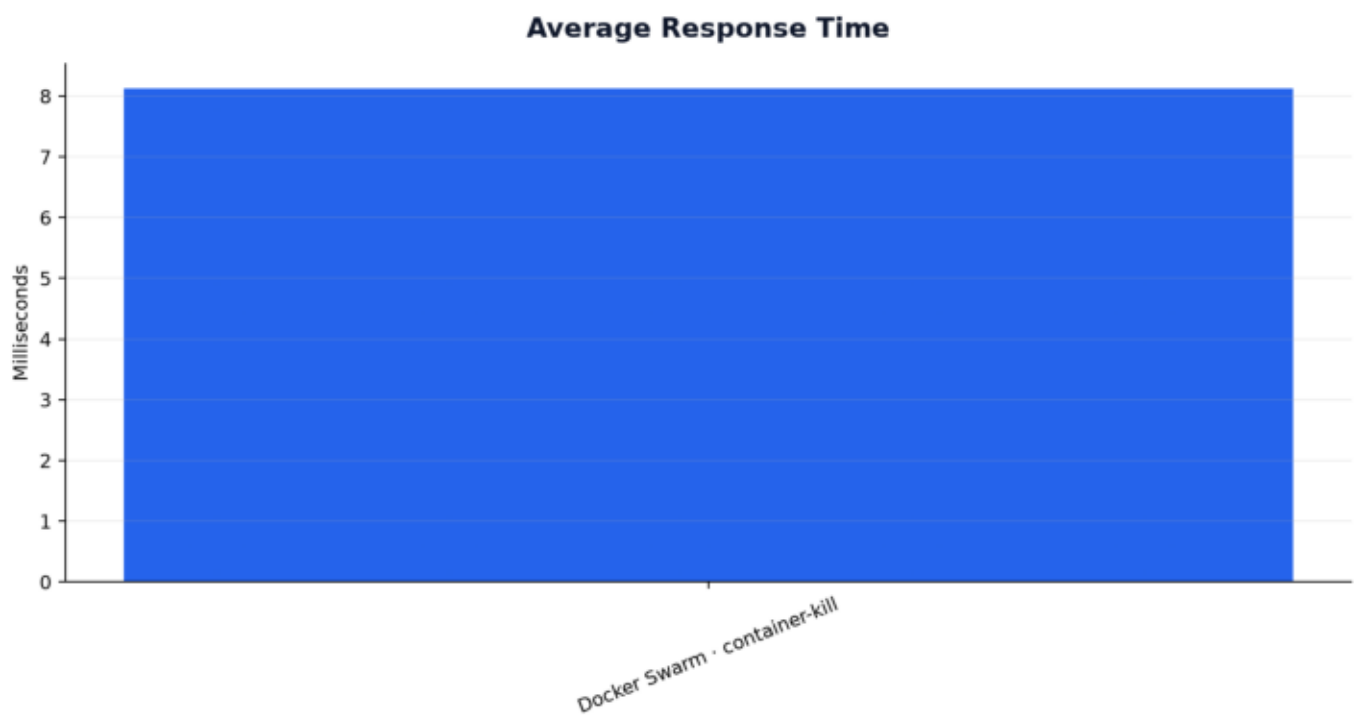
# Recovery Time

Current-session measured data



# Response Time

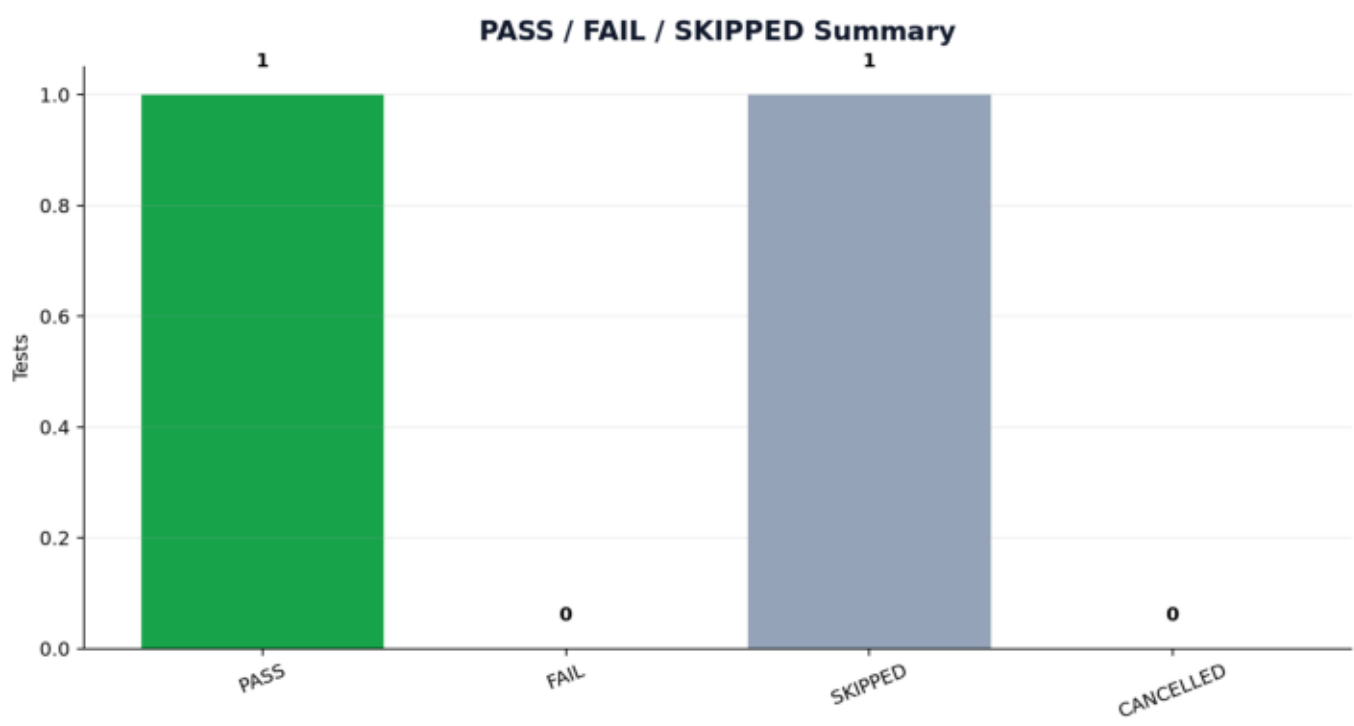
Current-session measured data





# Status Summary

Current-session measured data



# Failures and Errors

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| Platform | Scenario | Status | Error                 |
|----------|----------|--------|-----------------------|
| —        | —        | —      | No failures recorded. |

# Skipped Tests

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| Platform | Scenario     | Exact reason                                                        |
|----------|--------------|---------------------------------------------------------------------|
| swarm    | node-failure | Single-node Docker Swarm cannot demonstrate real node rescheduling. |

# Swarm vs Kubernetes Comparison

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| Platform | PASS | SKIPPED |
|----------|------|---------|
| swarm    | 1    | 1       |

# Final Conclusion

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During this campaign, 1 of 2 scenarios passed, 0 failed and 1 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.